Supervised Classification

Objectives:

- To generate supervised signatures using the AOI and Region Grow tools
- To use a signature file as an input to a maximum likelihood classification.

Study area:

- The image provided is a subset of a Landsat ETM+ image You are supposed to classify the image into five land use types:

Land cover/type	Code
Water	1
Roads	2
Residential	3
Forest	4
Vegetation	5

Instructions:

Classification is the process of **sorting pixels** into a finite number of individual classes, or **categories**, of data based on their data file values. If a pixel satisfies a certain set of criteria, then the pixel is assigned to the class that corresponds to that criteria. Open **pixel data** from **Meta data**. If I select value=50, name this pixel as water, the software will compute all matching DN values and will assign them to water class. There are two ways to classify pixels into different categories:

- Supervised
- Unsupervised

Supervised classification is more closely controlled by you than unsupervised classification. In this process, you select pixels that

represent patterns you recognize or can identify with help from other sources. Knowledge of the data, the classes desired, and the algorithm to be used is required before you begin selecting training samples. By identifying patterns in the imagery you can "train" the computer system to identify pixels with similar characteristics. By setting priorities to these classes, you supervise the classification of pixels as they are assigned to a class value. If the classification is accurate, then each resulting class corresponds to a pattern that you originally identified.

Unsupervised classification is more computer-automated. It allows you to specify parameters that the computer uses as guidelines to uncover statistical patterns in the data.

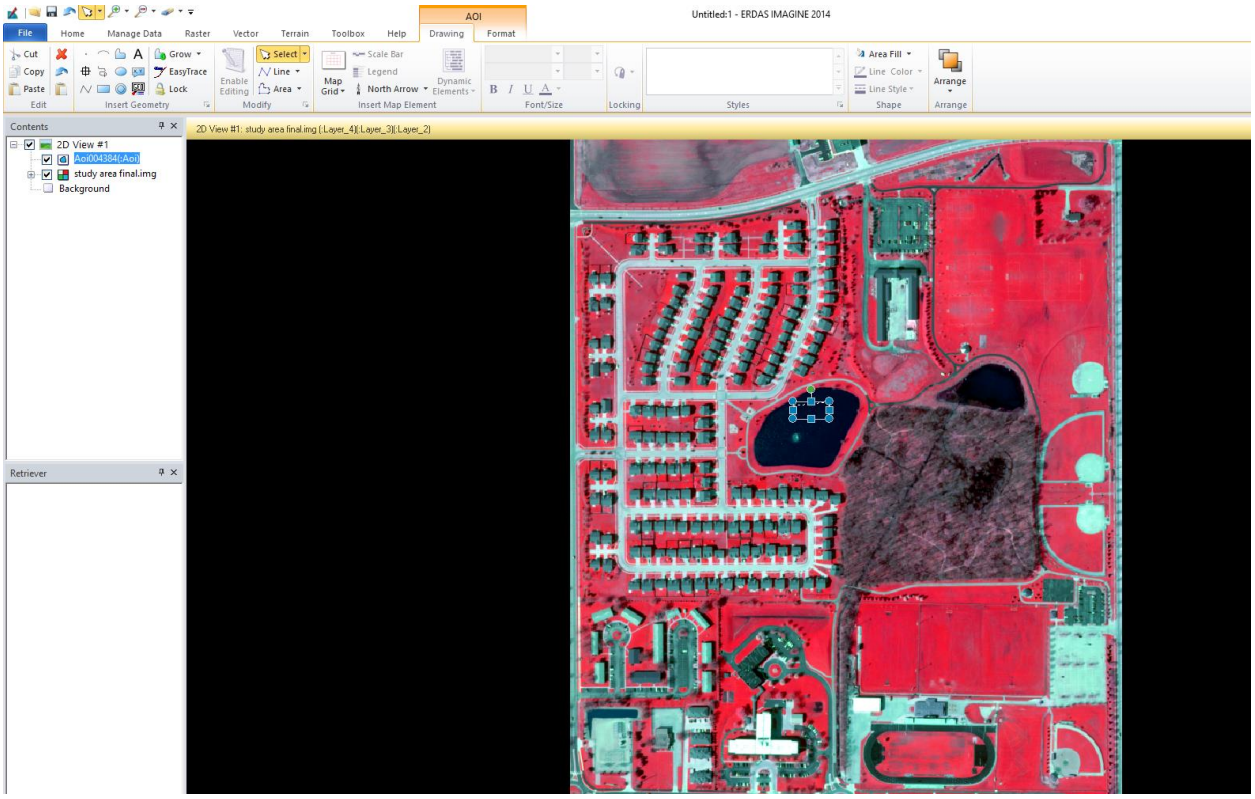
Supervised training requires a priori (already known) information about the data, such as: What type of classes need to be extracted? Soil type? Land use? Vegetation? What classes are most likely to be present in the data? That is, which types of land cover, soil, or vegetation (or whatever) are represented by the data?

In supervised training, the user relies on her/his own pattern recognition skills and a priori knowledge of the data to help the system determine the statistical criteria (signatures) for data classification. To select reliable samples, the user should know some information-either spatial or spectral-about the pixels that they want to classify. **The location of a specific characteristic, such as a land cover type, may be known through ground truthing.** Ground truthing refers to the acquisition of knowledge about the study area from field work, analysis or aerial photography, personal experience, etc. Ground truth data are considered to be the most accurate (true) data available about the area of study. They should be collected at the same time as the remotely sensed data, so that the data correspond as much as possible. However, some ground truth data may not be very accurate due to a number of errors, inaccuracies, and human shortcomings. Global positioning system receivers are useful tools to conduct ground truth studies and collect training sets. Training samples are sets of pixels that represent what is recognized as a discernible pattern, or potential class. The system will calculate statistics from the sample pixels to create a parametric signature for the class.

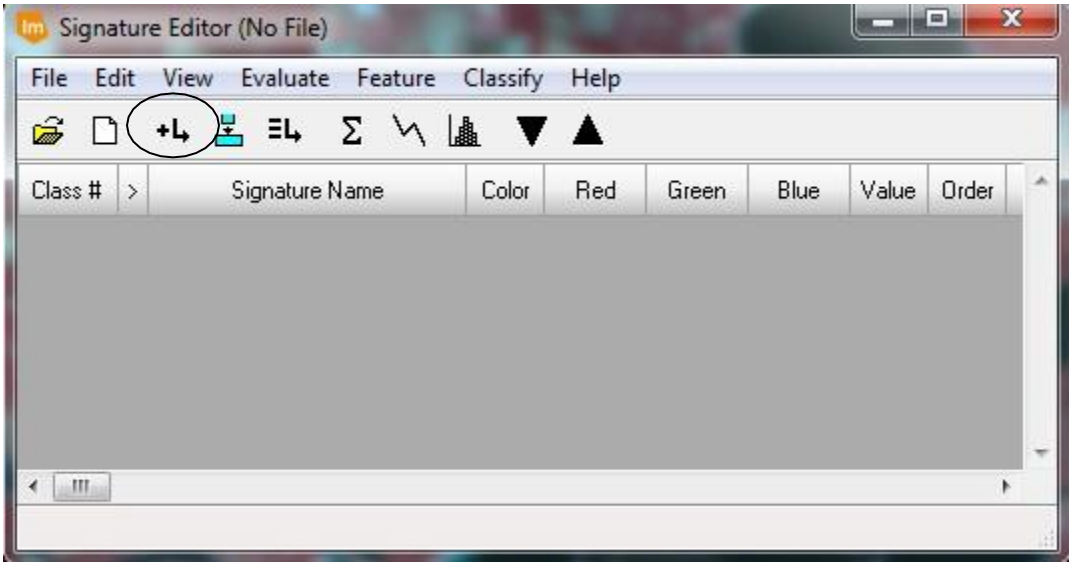
Procedure:

1. Start ERDAS Imagine, From the Viewer Tool Bar select the **Open Raster Layer** icon and select the file called study area final.img. Display as **Fit to Frame.**
2. From the Main Icon Panel select **Raster**-> **Supervise**-> **Signature Editor**. A dialog box will appear and will eventually contain a CellArray of created signatures.

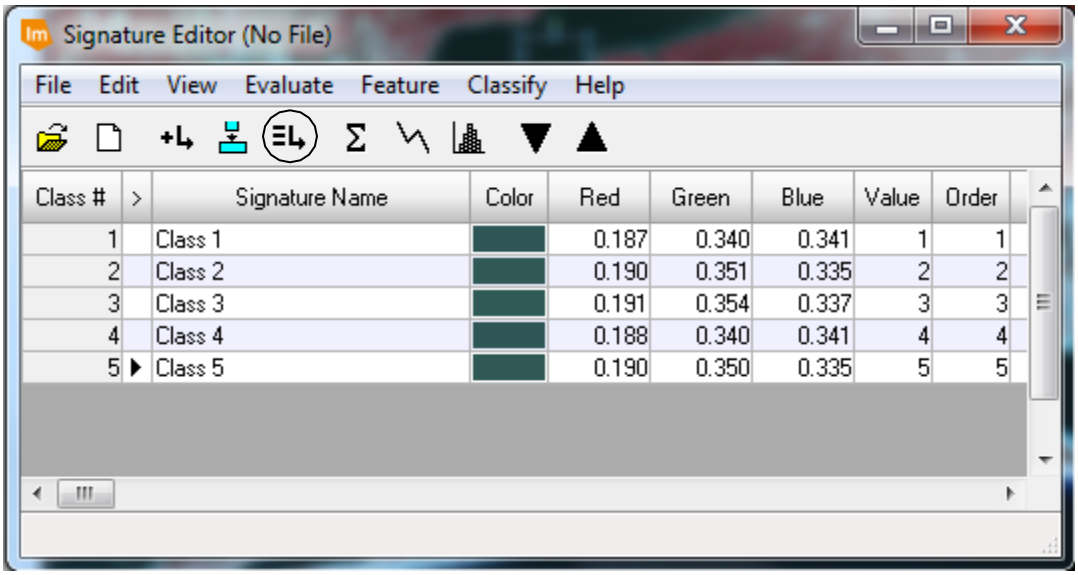
3. Go to drawing tab -> Draw a polygon -> as shown in figure I have taken a representative sample of the water.



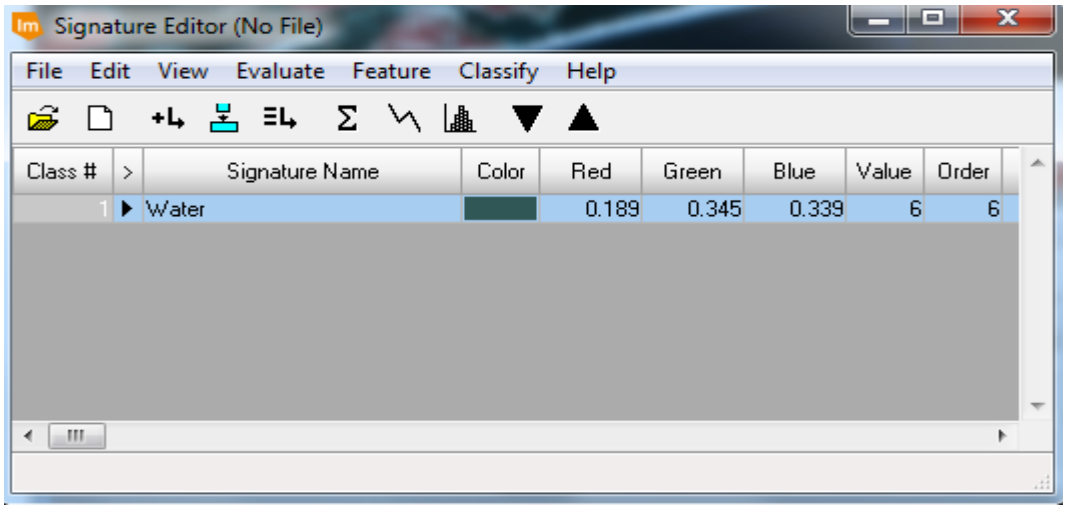
4. The area of the water will now be taken as a sample for the classifier. From the Signature Editor, Click Add Button, signature will appear in editor window.



5. Next take many samples of water all over the image and select all holding shift key as shown now click Add again.



6. Next select all signatures of water by click from 1-till last and click merge (circled above) an new class appears rename this to water and delete all others by right clicking and selecting Delete Selection.

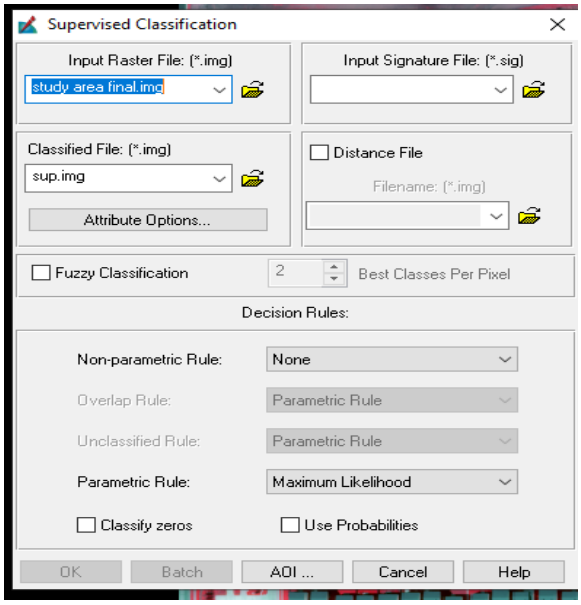


7. Repeat the process and add other land use classes.

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Water	1
Roads	2
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8. Once you are done From the Signature Editor Menu Bar select File/Save. Name your file,Click OK.

9. You may access the Supervised Classification box in one of two ways. Go to raster tab-> Supervised Classification. Specify inputs and outputs



10. For the Parametric Rule select Maximum Likelihood. If there are any signatures selected in the Signature Editor when you run your classification, then your output will be based only on that selection, so deselect any selected signatures. Use the remaining defaults and click OK.

11. When the process is done click OK in the Job Status box. Then display s, and the original image side by side in viewers opened using Geospatial Light Table option as Fit to Frame. Compare the two images.